Real-Time, 1-m Resolution Measurement of the Optical Distribution Network of a 21-km, 1:32 PON with a Coherent Optical Frequency Domain Reflectometer

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Abstract: We demonstrate a coherent optical frequency-domain reflectometer (C-OFDR) measuring a 21-km, 1:32 split passive optical network, achieving detailed Rayleigh-limited fiber characterization and identification of all distribution fiber ends with only 60-ms acquisition time.

1. Introduction

In recent years, coherent optical frequency-domain reflectometer (C-OFDR) research prototypes have been used for deep-ocean fiber sensing over trans-oceanic optical fiber communications systems. The speed and sensitivity of C-OFDR have enabled the first real-time measurements of seismic waves as they propagate over different subsea cable segments [1] and the first detection and tracking, with a fiber optic system, of a tsunami wave as it propagated more than 1000-km offshore [2]. These transoceanic fiber-optic systems have been engineered with "high-loss loopbacks" (HLLB) to enable optical time domain reflectometry (OTDR)-based loss and repeater gain measurements over an intentionally lossy testing loop that ensures sufficient optical isolation between high-speed optical communication wavelengths travelling in opposite directions. With measurement times of several hours, real-time monitoring was assumed to be impossible. The fact that C-OFDR operates in real-time over these intentionally lossy subsea test loops raises a compelling question: can C-OFDR be used for real-time monitoring of passive optical networks (PON), which have similarly high losses due to the presence of splitters with large splitting ratios?

In PON networks with large splitting ratios, conventional OTDR often faces limitations due to dynamic range constraints and the simultaneous desire for in-service network monitoring. While more advanced OTDR techniques have been used to enhance the dynamic range through coding and correlation [3], our work focuses on C-OFDR which offers unparalleled meter spatial resolution and high sensitivity with scan times on the order of tens of milliseconds.

Here we demonstrate the use of a C-OFDR to measure the optical distribution network of a PON with 21-km maximum reach and a 1:32 splitter. We show that all critical features are well resolved after 60 ms. This fast acquisition time indicates that a single C-OFDR can be used, in conjunction with an optical cross-connect switch, to monitor more than 10 PONs per second.

2. Experimental Setup

The experimental setup is shown in Fig. 1. The C-OFDR output and backscattered input are each amplified by an erbium-doped fiber amplifier (EDFA). The average power launched into the feeder for our measurements was 11.5 dBm. A circulator connects the two EDFAs to a 16-km feeder fiber, which is connected to a 1:32 splitter with a maximum loss of 16.9 dB. Downstream from the splitter, 32 distribution fibers, whose lengths are presented in Table 1, fan out to make up the PON. Of these, 12 are terminated with unpowered commercial transceivers to emulate the characteristics of a PON optical network unit (ONU), 4 are terminated with angle-polished connectors, and the rest are terminated by cutting the fiber, which results in a range of reflectivities that are at least several dB below the maximum 4% reflectivity typical of a properly cleaved fiber end.

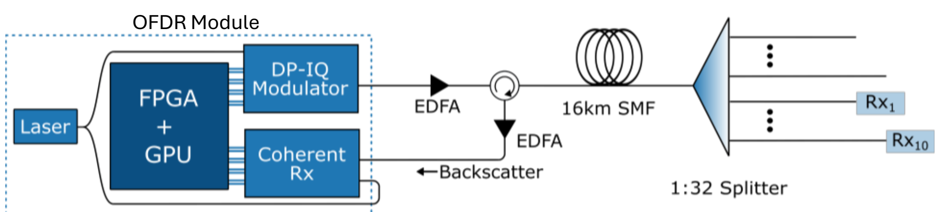

Fig. 1. Experimental setup

The C-OFDR module is constructed using conventional telecom coherent transceiver components, consisting of a dual polarization modulator and a coherent receiver. A 1552-nm laser with Hz-level linewidth serves as both the C-OFDR carrier and local oscillator. A single sideband dual polarization chirped pulse waveform with about 1-ms duration and pulse bandwidth of approximately 250 MHz was used. Details of the real-time OFDR DSP implemented using the combination of FPGA + GPU can be found in [1]. Post processing and block averaging 64 consecutive pulses were performed in real-time, resulting in about 1-m spatial resolution and 60-ms update time.

Table 1. Distribution fiber lengths (m)

Port	Fiber Length (m)	Port	Fiber Length (m)	Port	Fiber Length (m)	Port	Fiber Length (m)	Port	Fiber Length (m)	Port	Fiber Length (m)	Port	Fiber Length (m)	Port	Fiber Length (m)
1	41	5	96	9	181	13	230	17	270	21	341	25	1015	29	1250
2	52	6	111	10	191	14	241	18	280	22	391	26	1032	30	3015
3	73	7	151	11	200	15	250	19	291	23	433	27	1045	31	3350
4	81	8	161	12	220	16	260	20	311	24	462	28	1185	32	5000

3. Results

The trace of the entire PON, shown in Fig.2(a), was measured in 60-ms, a sufficient measurement period to reach Rayleigh-limited performance. The figure shows the 16-km feeder fiber followed by the 1:32 splitter and the different distribution fibers. The portions of the trace within the red and magenta boxes are shown in detail in (b) and (c) respectively.

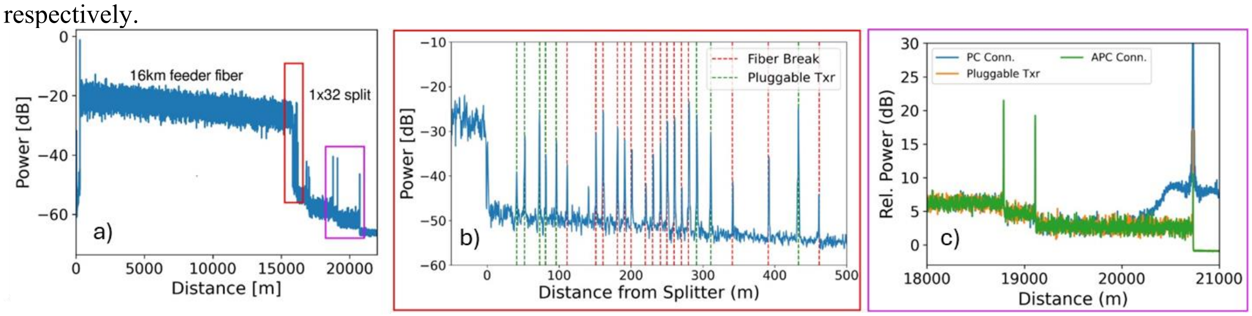

Fig. 2. C-OFDR traces with 60-ms averaging and 1-m spatial resolution. (a) shows a C-OFDR trace over the entire 21-km PON. The red and magenta rectangles are expanded to show (b) a 550-m section near the splitter, and (c) a detail over the last several km of the PON.

The red-boxed zoom-in Fig.2(b) is a 550-m segment starting approximately 50 meters before the 1:32 splitter, that includes reflection peaks from the first 24 distribution fibers ranging in length from 41 m to 462 m. Green dashed lines represent distribution fibers that are terminated with unpowered pluggable transceivers. Red dashed lines represent those terminated by breaking or cutting the fiber. All fiber ends, many spaced at 10-m intervals, were resolved. However, one small spurious peak located at approximately 140 m is not associated with a fiber end and may be due to a double reflection. All fiber spans exhibit strong random power variations, with the feeder section, as shown in the left trace (a) and the first 50 meters of the center trace (b), having larger peak-to-peak amplitude than the fibers after the splitter. This is the Rayleigh signature of the fiber, the fiber's fingerprint [4], which we will show in more detail in Fig. 3.

Importantly, these variations in power do not originate from random noise, but rather from Rayleigh scattering from fixed localized variations in the amorphous glass structure of the fiber. Thus, as multiple unique Rayleigh signatures from distinct distribution fibers are superimposed to the right of the splitter, the peak-to-peak variation decreases relative to the single feeder fiber. The zoom-in Fig.2(c) shows sections of the longest three distribution fibers with lengths 3015 m, 3350 m, and 5000 m. The two shorter spans are terminated with low-reflectivity angle-polished connectors (APC), and the termination of the 5000-m fiber is changed from APC (green trace), to a high-reflectivity PC connector (blue trace), to a pluggable transceiver (orange trace), highlighting the sensitivity and dynamic range of the C-OFDR. The pedestal in the blue trace, visible for approximately half a km on either side of the PC connector end, is caused by the strong PC reflection and limits the dynamic range and therefore the ability to resolve features close to the PC connector.

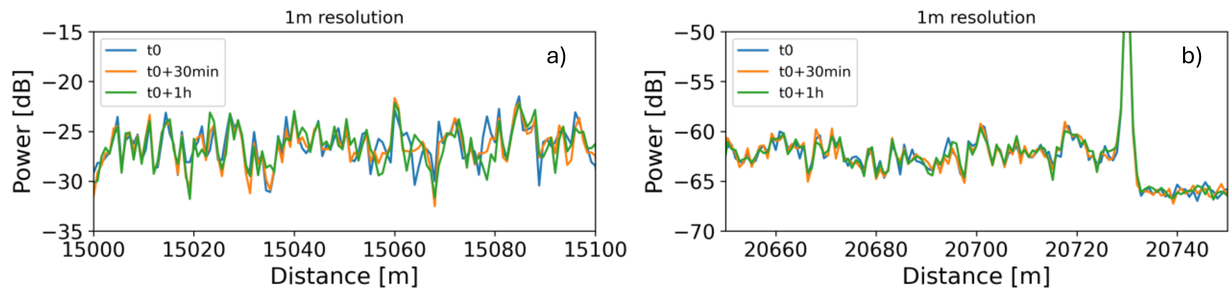

Fig. 3. (a) three traces show the Rayleigh backscattered signature along a 100-m length of feeder fiber taken at t_0 (blue trace), t_0+30 minutes (orange trace), and t_0+1 hour (green trace). (b) the same traces are displayed for the final 100 m of the longest distribution fiber, with the noise floor shown to the right of the reflection peak.

Fig. 3 shows three successive traces taken at time t_0, t_0+30 minutes, and t_0+1hour for (a) 100-m section of the feeder fiber and (b) a 100-m section at the end of the longest distribution fiber. The fine structure of the backscattered signal is relatively unchanged with time in both cases, indicative of the stability of the fiber's Rayleigh signature and that the SNR of a single OFDR pulse with 1-ms temporal duration is sufficient to fully resolve the signature. This data was also taken with 60-ms averaging.

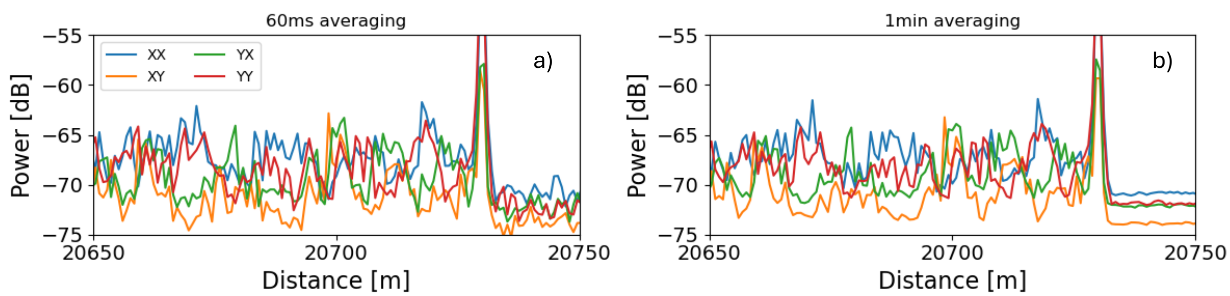

Fig. 4. Plots show all four components of the backscattered polarization around the reflection peak associated with the 5-km distribution fiber. The Rayleigh signatures for both (a) 60-ms averaging and (b) 1-minute averaging are equally well resolved.

The traces shown in Fig. 4 were taken with 60-ms averaging (a) and 1-minute averaging (b). The fact that the Rayleigh backscattered signatures in both cases are essentially identical indicates that there is no benefit to increasing the averaging time beyond 60-ms. The only significant difference between the two cases is the smoothing of the noise floor with 1-minute averaging.

4. Conclusion

We have demonstrated the performance of a coherent frequency-domain reflectometer over a 21-km PON with 32-way split. The C-OFDR, which was developed primarily for real-time monitoring of geophysical phenomena over subsea fiber cable systems, is also well suited for real-time PON monitoring. All fiber ends in our laboratory testbed were resolved, including multiple distribution fibers that differ in length by only 10 meters. The resolution of the C-OFDR is such that Rayleigh backscatter signatures were resolved beyond the splitter. The system reached Rayleigh-limited performance with an averaging time of 60-ms over our testbed, indicating that more than ten 1:32 PONs can be monitored per second with the addition of an optical cross-connect switch, or more than 600 PONs per minute. Although we have not yet tested the system over an in-service PON, the C-OFDR's operation over in-service trans-oceanic fiber optic systems, without penalty to the DWDM channels, indicates that operation over working PONs without adverse effects to PON traffic is feasible.

References

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